

AI-Driven Parametric CAD Generation

From Engineering Design Rules to Automated Fusion 360 Models

Review 2 — Working Prototype & Integration

Engineering Knowledge Base

Design rules, reference CAD

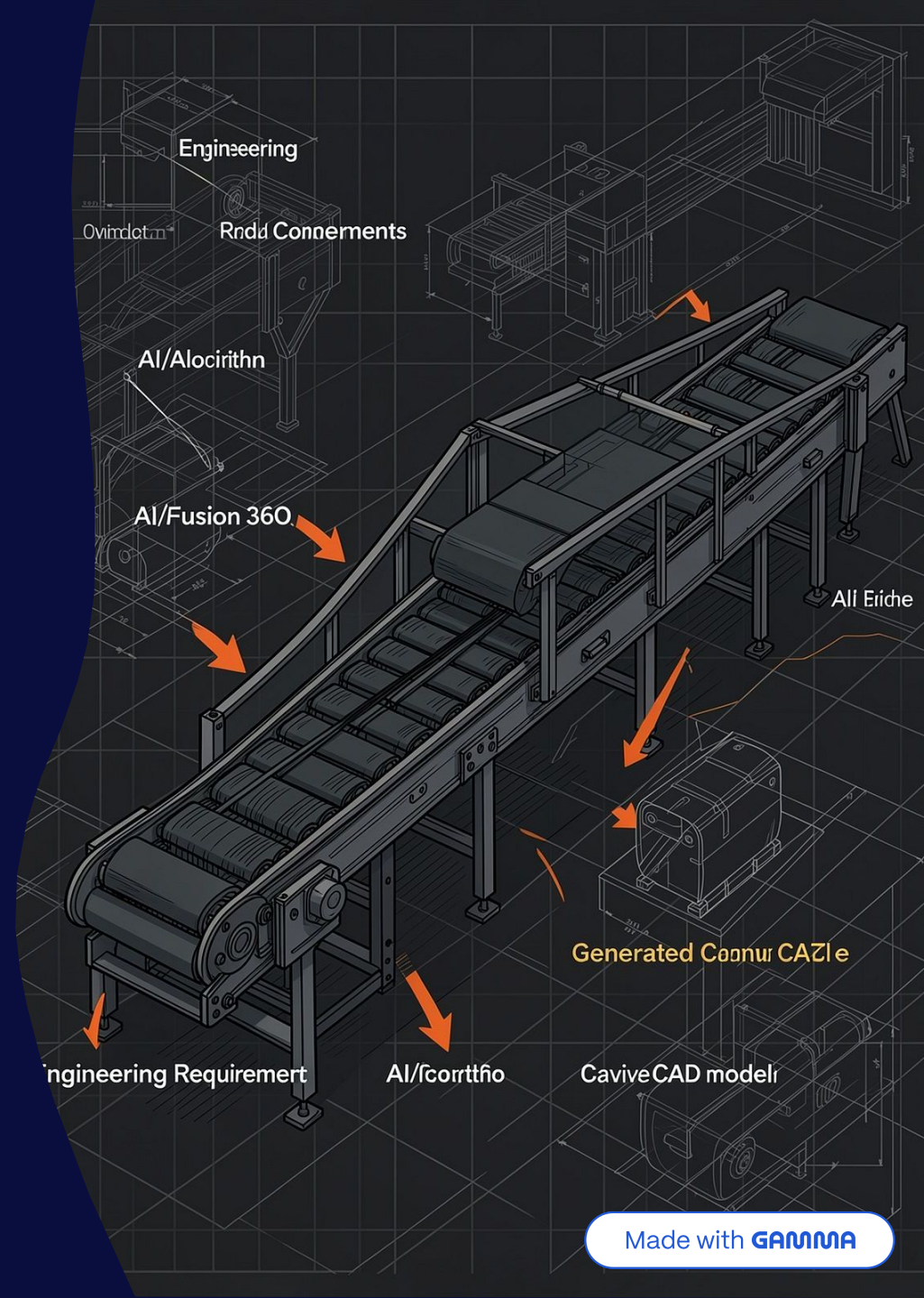
Parametric CAD Generation

Python + Fusion features

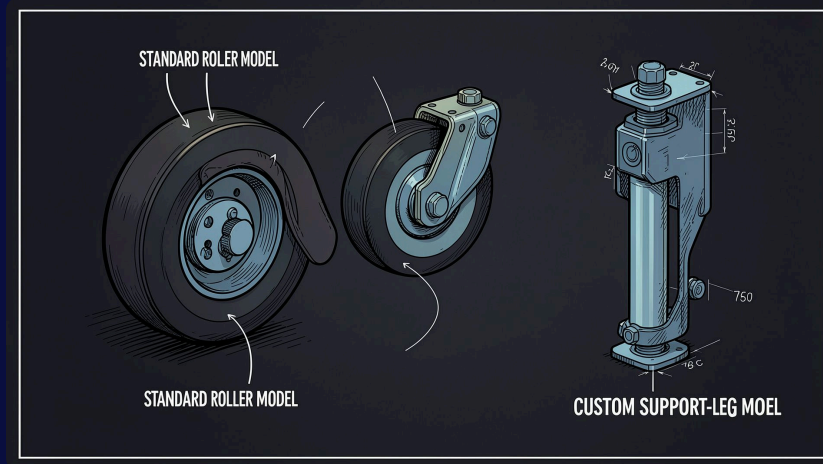
AI-Fusion Integration

Agent observes & edits Fusion

Current milestone: automated parametric conveyor CAD generation with AI-Fusion interaction. System remains a prototype — assembly automation and UI are next.



Mechanical Engineering → Computational Design Rules



Before automation the mechanical team encoded a standardized engineering knowledge base that constrains allowable geometry and ensures manufacturability.

Note: The algorithm is constrained by predefined rules and reference CAD — it does not freely invent mechanical geometry. Engineering documents act as the mechanical source of truth.

1

Conveyor dimensional limits

Max/min L, W, H

2

Roller diameter & pitch

Standard & tapered rollers

3

Frame & leg spacing

Support stations, deflection assumptions

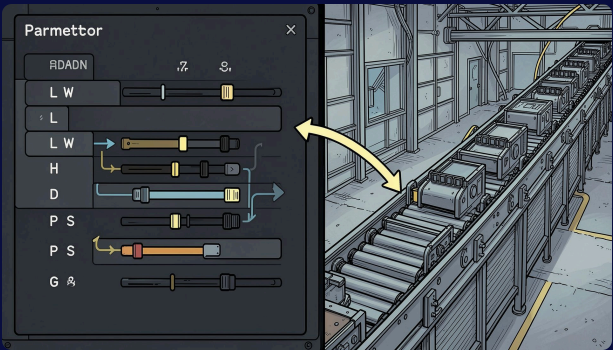
4

Guards & clearances

Safety margins, docking interfaces

From Engineering Parameters to Parametric Geometry

Parameters are defined once and drive parametric features so updates propagate through the model rather than forcing manual rebuilds.



Parameter table (abbreviated)

Parameter	Meaning
L	Conveyor length
W	Conveyor width
H	Frame height
D	Roller diameter
P	Roller pitch
S	Leg/support spacing
G	Guard height

Design principle: Parametric design = one source of truth for dimensions. Changing a master parameter updates dependent geometry.

L, W, H, D, P, S, G

Master dimensions and design inputs

Derived parameters

Roller count, leg count, margins

Parametric features

UserParameters → sketches → patterns → extrudes

AI-Python-Fusion 360 Architecture

The prototype stacks: an AI/design agent interacts with a Python generator which calls the Fusion 360 API to declare parametric features. Fusion's CAD kernel computes the B-rep.

01

Observation loop

- Agent captures screenshots / object tree → decides edits
- issues API actions

app → design → root → occurrence → component → features

02

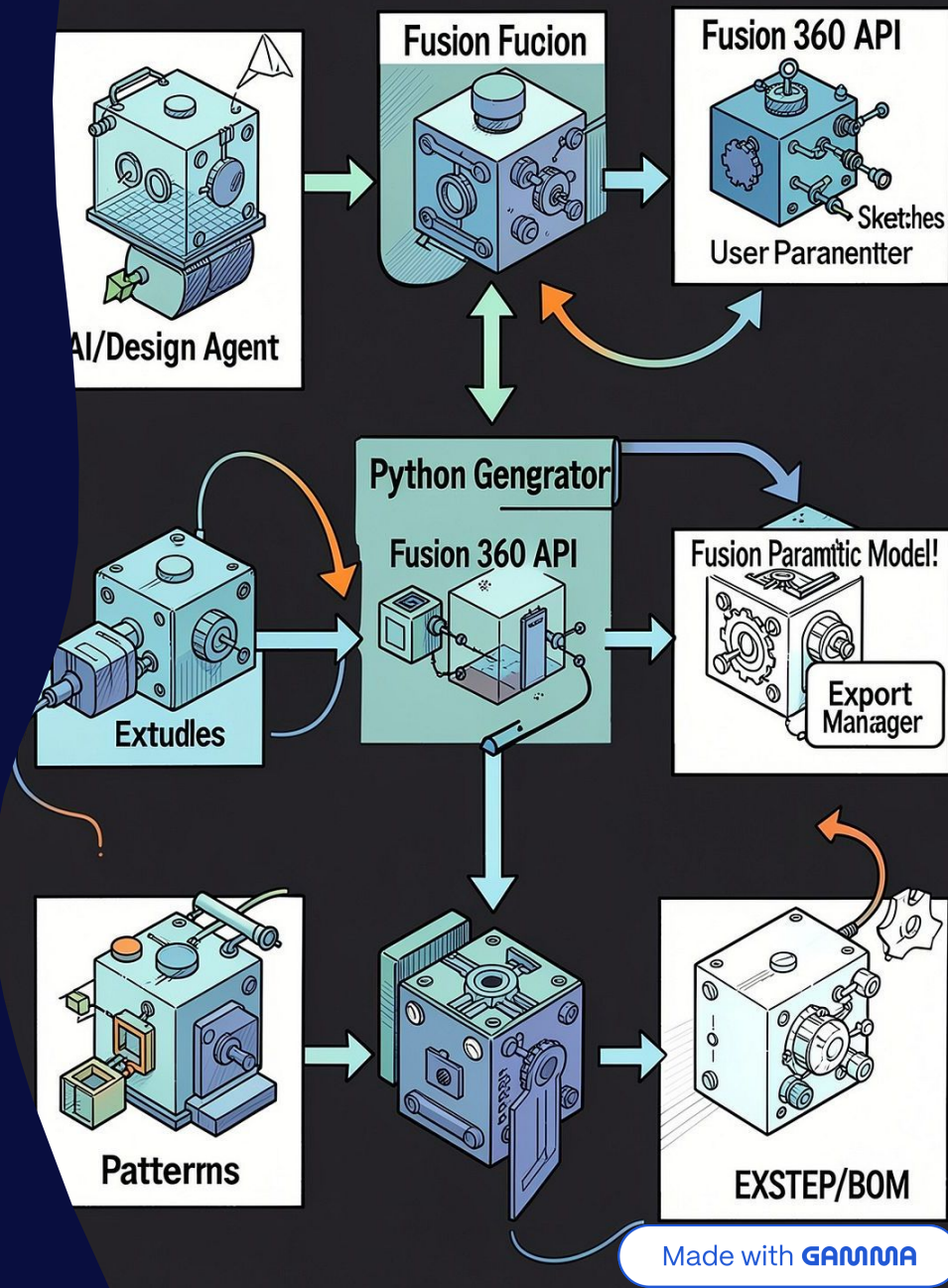
API primitives

adsk.core,
adsk.fusion,
UserParameters,
Sketches, Features

03

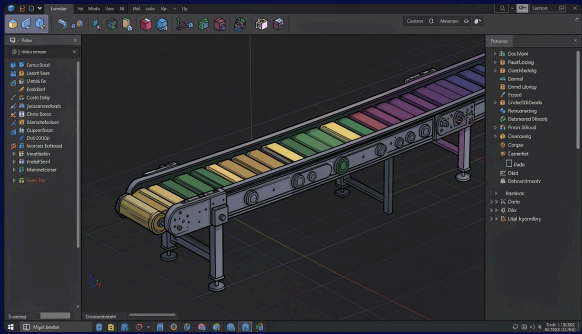
Export

STEP, BOM,
validation report



Current Working Pipeline

Notes: The pipeline includes offline unit tests of pure-Python logic and a mocked Fusion API for regression testing. Not all mechanical constraints are automated yet.



❑ Current result: automated parametric conveyor generation demonstrated inside Fusion 360.

- 1

1. INPUT PARAMETERS
Dimensions, roller spec, spacing, guards
- 2

2. ENGINEERING RULES
Range checks, counts, clearances
- 3

3. PARAMETRIC GENERATION
UserParameters + sketches + features + patterns
- 4

4. FUSION 360 SOLVE
CAD kernel computes B-rep
- 5

5. VALIDATION
Bounding box, counts, suppression state
- 6

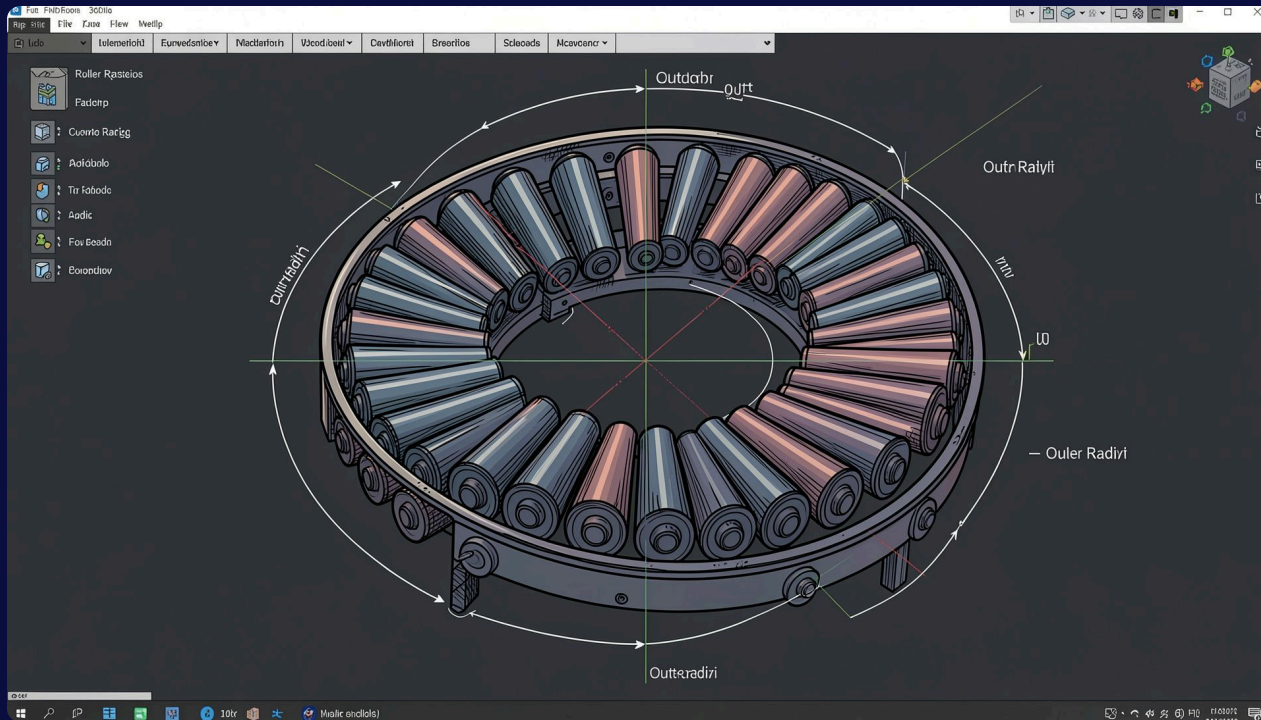
6. EXPORT
STEP + BOM + validation report

Extension to Curved Conveyor Geometry

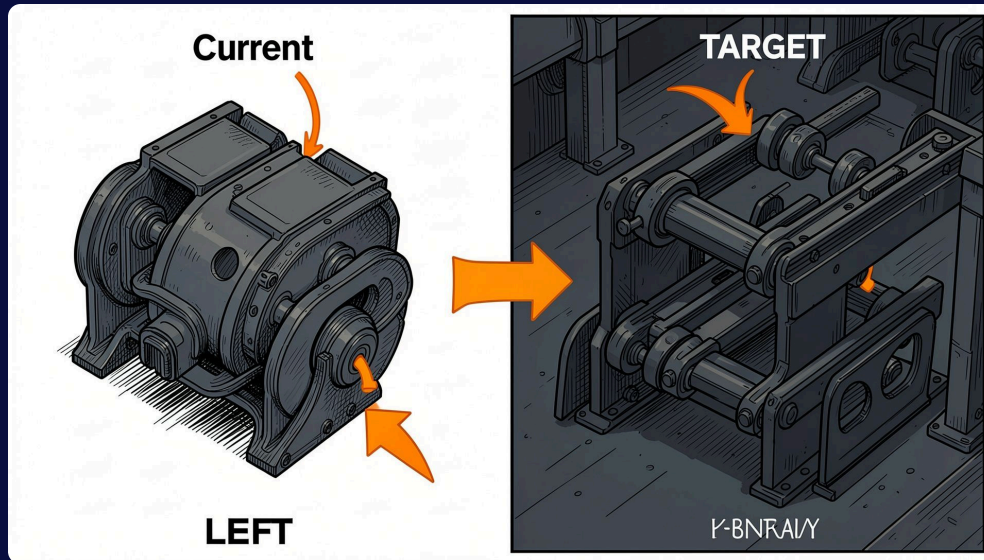
Curved Module — technical summary

- Curve angle, inner radius, conveyor width drive path
- Tapered roller geometry to match radial path
- Circular patterning for roller placement
- Inlet/outlet docking concepts for module connection

Implementation notes: The curve generator derives roller positions from polar coordinates and applies tapering to roller geometry. Demonstrated: 90° curved configuration generated and validated in development workflow. Not production-ready — further componentization required.



Current Limitation: Monolithic CAD → Target Mechanical Architecture



Current

- Generator produces consolidated geometry treated as one object
- Component identity not preserved

Target

- Component Generators → individual part files
- Assembly Generator → occurrences + joints + interfaces
- Meaningful BOM and replaceable components

❏ Next milestone: generate parts first, assemble second — this enables component reuse, editable part dimensions, and realistic manufacturing exports.

Next Architecture: Component-Based CAD Generation

Each component generator implements the reference CAD template and parametric instantiation rules. The assembly builder creates occurrences and defines joints at defined interface ports.



Roller Generator

Instantiate N rollers from reference geometry



Rail Generator

Extrude rails to length with cutouts for hardware

Docking ports/interfaces: standardized inlet/outlet ports allow straight and curved modules to be connected programmatically.



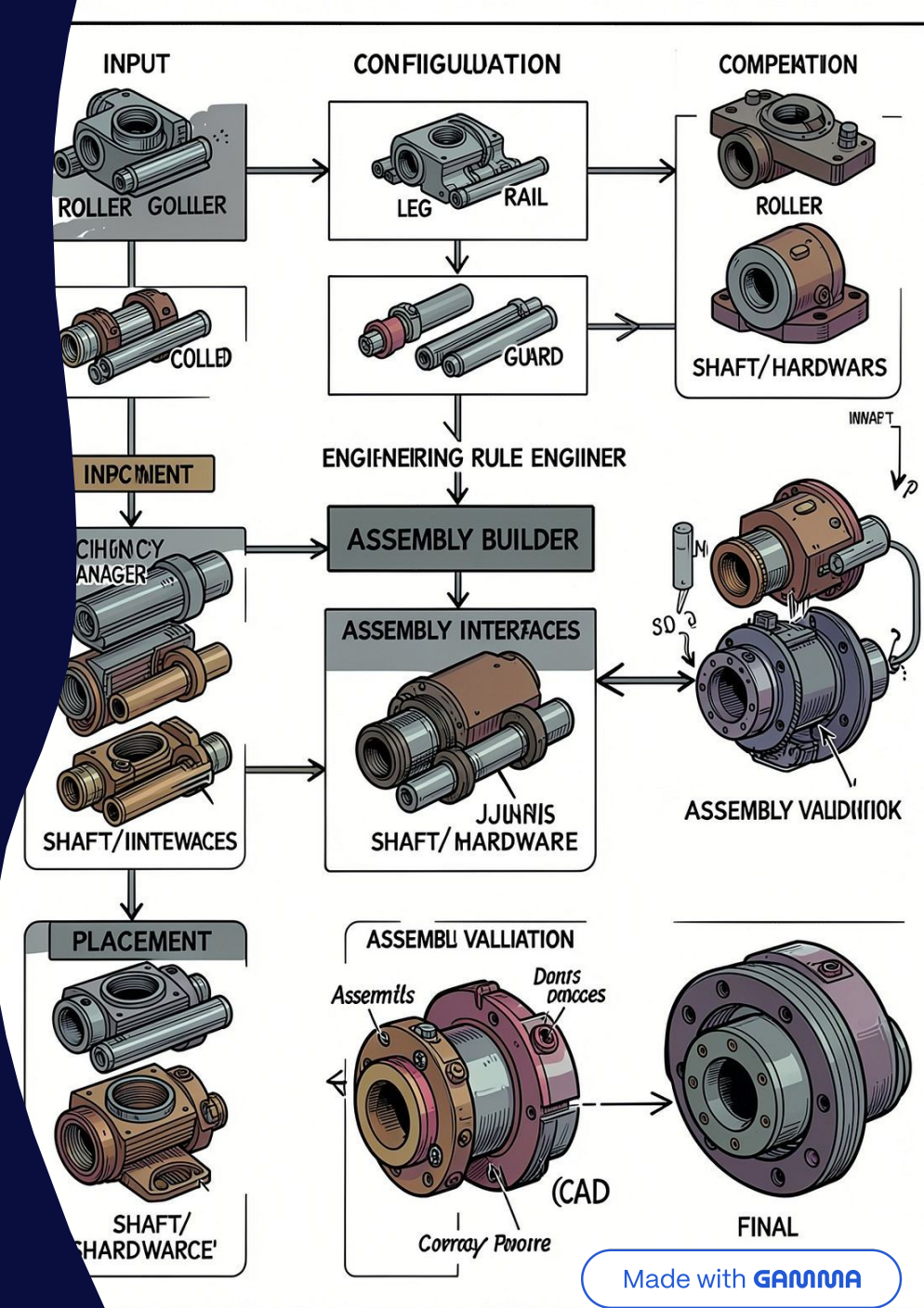
Leg Generator

Place supports based on span & load assumptions

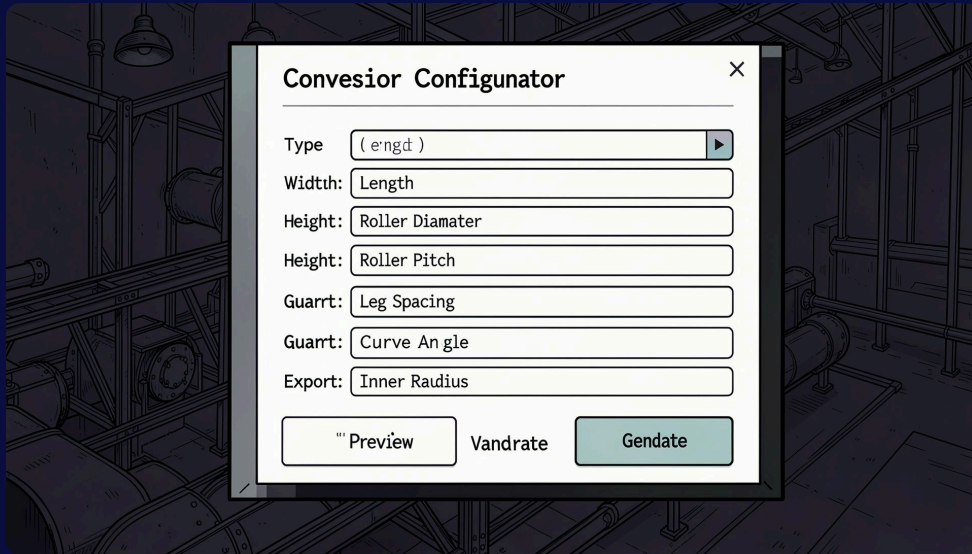


Guard Generator

Generate optional guards per parameters



User Interface & Validation



Current input: script-driven and developer-facing parameter entry (inputBox, JSON configs).

Target UI: persistent Fusion command dialog with typed CommandInputs, live preview, validation status and persistent logs.

Principle: Validation must be independent from visual inspection — automated checks gate exports.

Validation checks

Parameter ranges, roller count, leg count, bounding box

Geometry hygiene

Suppression state, intersecting solids, feature warnings

BOM consistency

Part IDs, standard hardware mapping, export readiness

Review 2 Outcome & Roadmap

COMPLETED

- Engineering rules documented
- Reference roller & leg CAD
- Parametric conveyor generator
- Fusion API integration
- AI-Fusion interaction demonstrated
- Curved module implemented

CURRENT

- Component-level CAD architecture
- Assembly generation strategy
- Component interfaces & docking
- Developer UI for configurator

NEXT

- Generate individual part files
- Automated assembly creation with joints
- Straight + curved module docking
- Replace inputBox with Fusion command UI

FINAL TARGET

AI-assisted engineering CAD generation that outputs structured assemblies, validated BOMs, and production-ready exports — achieved incrementally with human oversight.

Summary: Review 2 demonstrates the transition from engineering design rules to working automated CAD generation. Next milestone: move from consolidated geometry to component-aware assemblies and deploy a professional add-in UI for robust validation and module docking.

Current status: Working prototype — assembly automation and user interface under development.